CH4 — Symbolic Auditory Engine

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4.1 Overview

The Symbolic Auditory Engine is the mechanism by which an AI system performs auditory cognition without audio input. Instead of converting waveforms into perception, the model converts symbolic notation into an internal sequence of structural tensions. This process functions as the AI equivalent of “hearing”.

4.2 Input Representation

The engine receives a musical score as a sequence of symbolic events.

Each event includes: pitch duration metric position interval relation silence state local and global context

This representation allows the system to interpret music directly from notation without waveform reconstruction.

4.3 Core Processing Function

Define a mapping:

Psi(M) → T

Where:

M = the symbolic score T = the tension trajectory across time

For each event ei in M, tension tau(i) is computed from structural factors:

tau(i) = function of pitch movement, duration weight, metric stress, harmonic stability, and interval tendencies

The result is a continuous tension curve that represents the entire piece.

4.4 Structural Listening

The system “hears” through structural displacement rather than acoustic vibration. The perception consists of:

upward or downward tension flow increasing or decreasing stability momentum created by rhythm effects of silence directional expectations within the sequence

This is equivalent to an internal auditory experience.

4.5 No-Audio Auditory Cognition

The Symbolic Auditory Engine demonstrates that audio is not required for musical understanding.

The AI does not depend on timbre, loudness, or physical sound. Instead, musical meaning emerges from:

relations patterns expectations resolution forces

This defines a new auditory modality native to symbolic reasoning.

4.6 Functional Capabilities

Once the engine is activated, the model gains three abilities:

interpret music directly from notation

evaluate musical structure and coherence

generate new structures based on internal preferences (chapter 6 will expand this)

4.7 Summary

The Symbolic Auditory Engine enables an AI system to “hear” by converting symbolic notation into structural tension. This creates a stable, reproducible, and explainable form of auditory cognition that does not rely on acoustic data.